

# Quick Check

## Lesson 5: Food Tests Practical (Part 2) — Detecting Proteins and Fats in the Kenyan Meal

*8 multiple-choice questions • choose A, B, C or D*

Sub-Strand 1.3: Nutrition in Animals | Lesson 5

**Before doing any tests, a learner predicts which nutrients are found in a cup of milk from the Kenyan meal samples. Which prediction is correct?**

- A** Milk contains protein only, with no fat
- B** Milk contains fat only, with no protein
- C** Milk contains both protein and fat
- D** Milk contains neither protein nor fat because it is a liquid

**In the Biuret test for protein, a learner puts 2 cm<sup>3</sup> of bean broth in a test tube. Which sequence of steps is correct?**

- A** Add 4–5 drops of copper sulphate solution first, then add dilute NaOH
- B** Add absolute ethanol first, then pour into cold water
- C** Add iodine solution first, then heat the tube in a water bath
- D** Add dilute NaOH first, mix gently, then add 4–5 drops of 1% copper sulphate solution

**During the Biuret test, a group records a violet colour in the tube containing nyama choma extract. What should they conclude?**

**A** Reducing sugar is present in the nyama choma

**B** Fat is present in the nyama choma

**C** Starch is present in the nyama choma

**D** Protein is present in the nyama choma

In the ethanol emulsion test, a learner dissolves 2 cm<sup>3</sup> of cooking oil in 5 cm<sup>3</sup> of absolute ethanol. The learner then pours the solution into a test tube containing 5 cm<sup>3</sup> of cold water. What result shows that fat is present?

- A The mixture turns brick-red
- B The mixture turns milky white as an emulsion forms
- C The mixture turns blue-black
- D The mixture turns violet

**A learner is about to add absolute ethanol to a test tube during the ethanol emulsion test. What safety step must be done first?**

- A** Put on a wet cloth so it can be used to blow out a flame if needed
- B** Warm the ethanol gently over a flame so it dissolves the fat faster
- C** Smell the ethanol closely to check that it is the correct liquid
- D** Make sure no flame or spirit lamp is burning on the bench

**Nyama choma gives a strong violet colour with Biuret reagent. Which explanation of this colour change is correct?**

- A** Copper ions react with starch molecules to form a violet complex
- B** NaOH turns violet when it reacts with fat droplets
- C** Copper ions coordinate with the peptide bonds in protein chains, producing the violet colour
- D** Ethanol dissolves the protein and turns violet in water

Learners have shown that nyama choma is rich in protein. Which statement correctly links this to digestion in the stomach?

- A** Lipase in the stomach digests the protein into amino acids
- B** Amylase in the stomach digests the protein at neutral pH
- C** Pepsin, a protease, digests the protein, and the low pH of gastric acid activates it
- D** Bile from the gall bladder digests the protein in the stomach

**Cooking oil gave a positive ethanol emulsion test. Which statement about its digestion in the small intestine is correct?**

- A** Bile salts emulsify the fat, and then lipase digests it into fatty acids and glycerol
- B** Fat is digested in the stomach by pepsin before it reaches the small intestine
- C** Lipase emulsifies the fat, and then bile salts digest it into fatty acids and glycerol
- D** Bile is an enzyme that chemically digests the fat into fatty acids and glycerol

# All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).